

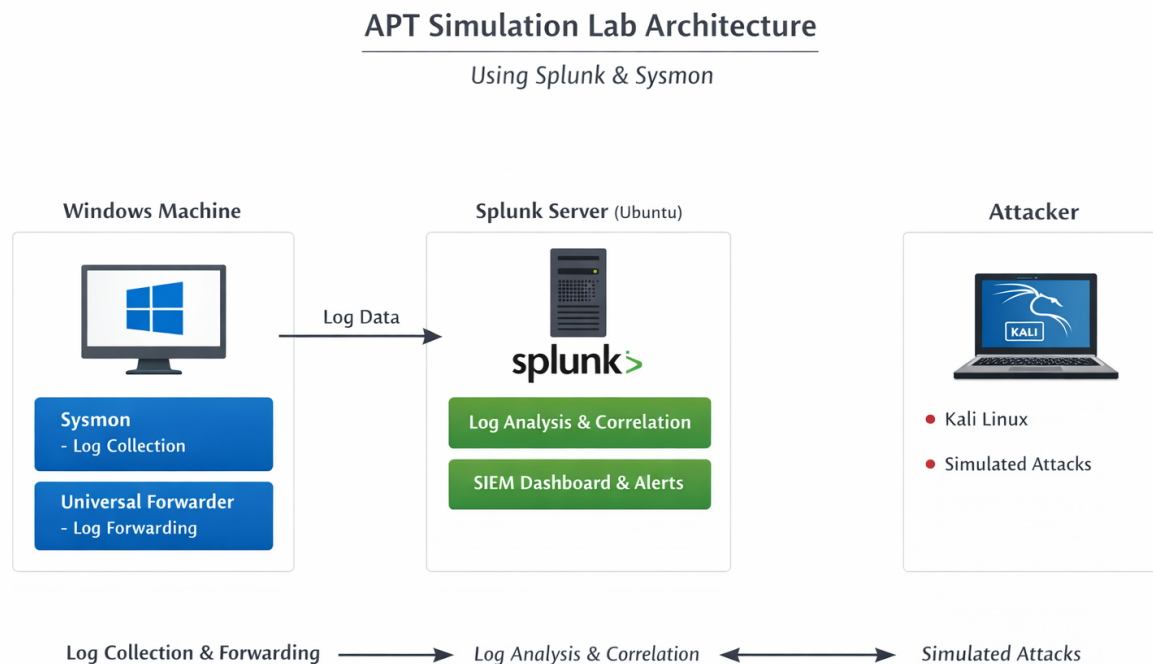
System Architecture

Overview

The architecture of this project is designed to simulate a real-world **Security Operations Center (SOC)** environment, where system activities are monitored, collected, and analyzed to detect potential threats.

The system follows a centralized logging and detection approach using **Sysmon**, **Splunk Universal Forwarder**, and **Splunk Server**.

Architecture Diagram



Components

1. Windows Machine (Target System)

- Acts as the primary system where activities are generated
- Sysmon is installed to capture detailed logs

- Generates:
 - Process creation logs (Event ID 1)
 - Network connection logs (Event ID 3)
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2. Sysmon (System Monitor)

- Captures detailed system-level events
 - Tracks:
 - Process execution
 - Command-line activity
 - Network connections
 - Provides enhanced visibility beyond default Windows logs
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3. Splunk Universal Forwarder

- Installed on the Windows machine
 - Collects logs from:
 - Sysmon
 - Windows Event Logs
 - Forwards logs to Splunk Server
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4. Splunk Server (Ubuntu)

- Central log collection and analysis system
 - Receives logs via port **9997**
 - Provides:
 - Search and analysis
 - Detection using SPL queries
 - Alert generation
 - Dashboard visualization
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5. Dashboard & Alerting Layer

- Displays real-time system activity
 - Shows:
 - PowerShell usage
 - Network activity
 - Process execution
 - Generates alerts for suspicious behavior
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Data Flow (Step-by-Step)

1. User or attacker executes a command (CMD / PowerShell)
 2. Sysmon captures the activity and generates logs
 3. Logs are collected by the Universal Forwarder
 4. Forwarder sends logs to Splunk Server (Port 9997)
 5. Splunk indexes and stores the logs
 6. Detection rules (SPL queries) analyze the logs
 7. Alerts are triggered if suspicious activity is found
 8. Dashboard visualizes the detected events
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Network Configuration

Component	IP Address	Purpose
Windows Machine	192.168.42.128	Log generation
Splunk Server	192.168.42.132	Log analysis
Port 9997	Forwarding	Log transfer
Port 8000	Web Interface	Splunk UI

Key Design Principles

- **Centralized Logging** → All logs are collected in one place
 - **Real-Time Monitoring** → Immediate visibility into activities
 - **Detection-Based Security** → Focus on identifying threats
 - **Scalability** → Can be extended to multiple systems
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Security Considerations

- The entire setup runs in an isolated lab environment
- No external systems are affected
- Used only for educational and ethical purposes